16 December 1969
SUPERSEDING
MIL-I-11519C
17 September 1963
(see 6.3)

MILITARY SPECIFICATION

INTRENCHING TOOL, HAND, FOLDING, LIGHTWEIGHT, AND CARRIER

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE AND CLASSIFICATION

1.1 This specification covers one size, lightweight, collapsible, hand intrenching tool and nylon carrier (see 6.1).

1.2 Classification.- The intrenching tool shall be of the following types, as specified (see 6.2).

Type I - With carrier, nylon

Type II - Without carrier

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-A-200/13	-	Aluminum Alloy Bar, Rod, Shapes, and Tube, and Wire, Extruded, 7178
QQ-A-250/12	-	Aluminum Alloy 7075, Plate and Sheet
QQ-A-591	-	Aluminum Alloy Die Castings
QQ-W-423	-	Wire, Steel, Corrosion-Resisting
TT-C-490	-	Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings

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TT-E-529	-	Enamel, Alkyd, Semi-Gloss
WW-T-700/3	-	Tube, Aluminum Alloy, Drawn, Seamless 2024
PPP-B-601	-	Boxes, Wood, Cleated-Plywood
PPP-B-636	-	Box, Fiberboard
PPP-C-843	-	Cushioning, Material, Cellulosic
PPP-P-291	-	Paperboard, Wrapping, Cushioning
PPP-T-76	-	Tape, Pressure-Sensitive Adhesive Paper (for Carton Sealing)

MILITARY

MIL-H-6088	-	Heat Treatment of Aluminum Alloys
MIL-A-8625	-	Anodic Coatings, For Aluminum and Aluminum Alloys
MIL-C-43683	-	Carrier, Intrenching Tool, Hand, Folding, Lightweight

STANDARDS

FEDERAL

FED-STD-66	-	Steel: Chemical Composition and Hardenability
FED-STD-595	-	Colors

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-130	-	Identification Marking of U.S. Military Property

DRAWINGS

ARMY NATICK LABORATORIES

2-2-284	-	Intrenching Tool, Hand, Folding, Lightweight, Assembly Complete
2-2-285	-	Intrenching Tool, Hand, Folding, Lightweight; Details A
2-2-286	-	Intrenching Tool, Hand, Folding, Lightweight; Details B
2-2-288	-	Intrenching Tool, Hand, Folding, Lightweight, Strength Test

(Miniature copies of Drawings 2-2-284, 2-2-285, 2-2-286, and 2-2-288, identified as figures 1, 2, 3, and 4 respectively, and figure 5 are attached for information purposes only.)

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(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issues in effect on date of invitation for bids or request for proposal shall apply:

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

E 18-65 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to American Trucking Associations, Attn: Tariff Order Section, 1616 P Street, N.W., Washington, D.C. 20036.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 202 Union Station, 516 W. Jackson Blvd, Chicago, Illinois 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article.- Unless otherwise specified (see 6.2), before production is commenced, a preproduction sample, consisting of a finished intrenching tool with or without carrier, as applicable, covered by this specification, shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.2. The approval of the preproduction sample authorizes the commencement of production, but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production items.

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3.2 Alternate components.- Components offered as equivalent to components specified herein and on the referenced drawings as a specific manufacturer's part or model number or equal shall be functionally equal to the manufacturer's part or model number identified and of equal or better quality. The incorporation and inclusion of such a component in the end product shall not require modification or change to any other specified component, and shall not reduce ease of maintenance to it or any other component, unless such change is specifically approved by the contracting officer. Prior to manufacture of the preproduction sample, or, if none, the initial production unit, the supplier shall submit, for the contracting officer's approval, a list identifying each proposed alternate component, together with proof that each listed component complies with requirements above. The contracting officer, at his option, may require a physical sample of the proposed substitution. Approval of the submitted listing, together with necessary supporting data, authorizes commencement of manufacture of the preproduction sample or initial production unit, as applicable, but does not relieve the supplier of the responsibility that these components perform in accordance with specified requirements when incorporated into the end product.

3.3 Materials.- Materials not definitely specified shall be of the quality normally used by the manufacturer for intrenching tools and for intrenching tool carriers, provided the completed items comply with all the provisions of this specification.

3.3.1 Tube aluminum alloy drawn, seamless, 2024.- The drawn aluminum alloy tube shall conform to 2024-0 and 2-24-T42, as applicable, of WW-T-700/3.

3.3.2 Aluminum alloy tube, cold drawn, seamless, 7075.- The aluminum tube shall be commercial, cold drawn, seamless, 7075-T6.

3.3.3 Aluminum alloy tube, 7178.- The aluminum alloy tube shall conform to 7178-0 of QQ-A-200/13.

3.3.4 Aluminum alloy plate and sheet, 7075.- The aluminum alloy plate shall conform to 7075-T6 of QQ-A-250/12.

3.3.5 Aluminum alloy die castings.- The aluminum alloy die castings shall conform to alloy No. A380 of QQ-A-591.

3.3.6 Lubricant.- The commercial type lubricant shall be a solution of a waxlike silicone compound in a nonflammable solvent. The lubricant shall be an extreme pressure type, color-free, and nonstaining.

3.3.7 Enamel.- The enamel shall conform to class B of TT-E-529. The color shall be black.

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3.3.8 Steel.- The steel shall be as specified on Drawing Nos. 2-2-285 and 2-2-286 and shall conform to the chemical composition of FED-STD-66.

3.3.9 Wire, steel, corrosion-resisting.- The corrosion-resisting steel wire shall conform to form 1, composition 302, condition B of QQ-W-423.

3.4 Design and construction.- The intrenching tools shall conform to the design, details, dimensions and materials specified on Drawing No. 2-2-284, 2-2-285, 2-2-286, and 2-2-288, and as specified herein. The blade portion of a finished intrenching tool shall move on the hinges to picking and shoveling positions when tested as specified in 4.4.1. The intrenching tool shall show no fracture of components, and shall have no permanent deflection to exceed 1/4 inch when tested as specified in 4.4.2. The hinge and the rear handle shall bear evenly on the thrust washer and thrust plate, respectively, in operating position.

3.4.1 Rivets, anti-rotation insert, washer, hinges and blade.- Rivets insert, washer, hinges and blade shall be fabricated of the material specified on applicable drawings.

3.4.2 Riveting.-

3.4.2.1 Hinge to blade.- Hinge shall be riveted to the blade. When the solid rivets are driven, they shall completely fill the holes. The rivet heads shall be full, neatly made, concentric with the rivet hole, and shall be in full contact with the surface of the riveted member.

3.4.2.2 Semi-tubular rivets.- The end of the rivets shall be neatly mushroomed and shall be concentric with the rivet holes. The compression of the rivets shall be controlled to meet the test as specified in 4.4.1.

3.4.3 Heat treatment.-

3.4.3.1 Steel components.- Steel components, except for rivets and spring, shall be heat treated as shown on applicable drawings. Hardness shall be determined in accordance with 4.4.3.

3.4.3.2 Tension tube and rear handle.- Tube and rear handle shall be heat treated to the tempers on specified applicable drawings and in accordance with MIL-H-6088.

3.4.4 Operating spring.- The operating spring shall be fabricated of the material specified in 3.3.9.

3.4.5 Carrier.- The carrier for the type I intrenching tool shall be fabricated in accordance with MIL-C-43683.

3.4.6 Type I intrenching tool.- The type I intrenching tool shall be as illustrated on figure 5. Figure 5 is included for information only.

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3.5 Finish.-

3.5.1 Steel components.- Prior to assembly, the steel components shall be pretreated for painting in accordance with any method or type of TT-C-490.

3.5.1.1 Enamel.- After pretreatment, the steel components, except thrust washer, shall be coated with the enamel specified in 3.3.7.

3.5.2 Aluminum components.- Prior to assembly, the aluminum components, except spring pin, shall be anodized in accordance with type II, class 2 of MIL-A-8625. The color shall be No. 37038 of FED-STD-595.

3.5.3 Pre-bonding preparation.- Prior to the bonding specified in 3.5.3.1, the bonding surfaces of the thrust washer shall be sand blasted. The thrust washer and nut shall be thoroughly cleaned of dust, dirt and grease.

3.5.3.1 Bonding.- The thrust washer and nut shall be bonded with the adhesive specified on Drawing No. 2-2-285. The adhesive shall be applied in the manner recommended by the manufacturer. After assembly and bonding, the thrust washer and nut shall be given a coat of enamel specified in 3.3.7.

3.6 Lubrication.- Prior to assembly of nut to handle, the threaded portions of the nut and handle shall be given a light coat of the lubricant specified in 3.3.6. The hinge areas shall be coated with the lubricant specified in 3.3.6.

3.7 Marking for identification.-

3.7.1 Intrenching tool.- The intrenching tool shall be marked for identification as shown on Drawing No. 2-2-285 and applied in accordance with MIL-STD-130.

3.8 Workmanship.- The surface of the intrenching tools shall be free of burrs, slivers, sharp corners, and rough or sharp edges, except for the cutting edges of the blade. The feather edge shall be removed from the beveled, serrated cutting edges. The coating shall be continuous, uniform, smooth, dry film without area of no film, and shall be free from embedded foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves

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the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.- When a preproduction sample is required, it shall be examined for defects as specified in table II, dimensions specified, and tested as specified in 4.4.1 and 4.4.2. The presence of any visual defect, any dimensions not within the specified requirements, or failure to pass the tests shall be cause for rejection of the preproduction sample.

4.3 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase documents.

4.3.1.1 Intermediate testing.- Prior to assembly, the blade, hinges and thrust washer shall be tested in accordance with 4.4.3 to determine compliance with 3.4.3.1. The hinges, thrust washers, and blades shall be tested separately. The lot shall be all the blades, hinges or thrust washers, as applicable, heat treated as a group and offered for inspection at one time. The sample unit shall be one blade, one hinge or one thrust washer. The inspection level shall be S-2 with an acceptable quality level (AQL) of 2.5 defects, expressed in terms of defects per hundred units.

4.3.1.2 Process examination.- Examination shall be made during the manufacturing process for the requirements specified in table I to establish that no deviation is made from indicated requirements. Whenever nonconformance is noted, correction shall be made to affected items and process.

TABLE I.- Process Inspection

Requirements	Requirement paragraph
Rivet holes filled	3.4.2.1
Pretreatment of steel components prior to painting	3.5.1
Pre-bonding preparation	3.5.3

4.3.2 End item inspection.- The inspection lot shall be all complete intrenching tools of one type offered for inspection at one time. The sample unit shall be one intrenching tool; with or without carrier, as applicable.

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4.3.3 Visual examination.- Visual examination of each sample unit shall be made for the defects in table II. The inspection level shall be II with an AQL of 2.5 for major defects, and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE II.- Classification of defects

Examine	Defect	Classification Major	Minor
Intrenching tool			
Finish	Not type specified	X	
	Color not as specified		X
	Area of no paint		X
	Contains streaks, runs, drops, grit or wrinkles		X
	Dirt or other foreign matter imbedded in finish		X
Construction	No lubricant on threads of nut and handle	X	
	Component not type specified	X	
	Carrier missing (type I)	X	
	Hinge does not bear evenly on thrust washer in operating postion	X	
	Rear handle does not bear evenly on thrust plate in operating postion	X	
	Handle assembly off center	X	
	Serrated and bevel edge missing	X	
	Edges not free from burrs, slivers and sharp corners, except beveled, serrated cutting edges	X	
	Feather edge not removed from beveled, serrated cutting edges		X
	Any rivet missing	X	
	Rivet not peened, curled, or end not mushroomed	X	
	Rivet head missing	X	
	Any rivet reversed, e.g., head is not on the intended side of component as required		X
	Rivets not in full contact with surface of riveted member	X	
	Threads on locking nut and outer handle crossed or damaged	X	
Identification marking	Omitted, incorrect, illegible, misplace, or size of characters, not as specified		X

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4.3.4 Dimensional examination.- Examination shall be made of the intrenching tool for compliance with dimensions specified. Any dimension not meeting specified requirements shall be classified as a defect. The inspection level shall be S-3 with an AQL of 1.5 defects, expressed in terms of defects per hundred units.

4.3.5 End item testing.- Sample units of the intrenching tool shall be tested in accordance with 4.4.1 and 4.4.2. The inspection level shall be S-2 with an AQL of 2.5 defects, expressed in terms of defects per hundred units.

4.3.6 Examination of preparation for delivery.- An examination shall be made to determine that packaging, packing and marking as required by section 5 of this specification are complied with. Defects shall be as indicated in table III. The sample unit shall be one shipping container fully prepared for delivery. The lot shall be all of the containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0, expressed in terms of defects per hundred units.

TABLE III.- Examination of preparation for delivery

Examination	Defect
Markings	Omitted; incorrect, illegible; of improper size, location, sequence, or method of application
Materials	Component missing or damaged
Workmanship	Inadequate application of components, such as cushioning inadequate or bulging or distortion of interior boxes
Contents	Number per interior box and shipping container is more or less than required

4.3.6.1 Examination of shipping containers.- When shipping containers are required to be in accordance with PPP-B-601, examination for defects in closure and strapping shall be in accordance with the appendix of that specification.

4.4 Tests.-

4.4.1 Function.- With the nut disengaged and the handle held firmly, the blade shall fold under loads of not less than 2 pounds and not more than 12 pounds. The load shall be applied perpendicular to the face of the blade and 1/2 inch from the tip of the blade. Noncompliance with 3.4 and 3.4.2.2 constitutes failure of this test.

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4.4.2 Strength tests.- The intrenching tool shall be tested in accordance with methods and procedures specified on Drawing No. 2-2-288. Noncompliance with 3.4 shall constitute a defect.

4.4.3 Hardness.- Hardness tests to determine compliance with 3.4.3.1 shall be in accordance with ASTM E 18-65. Three readings shall be made on the blade within a 4 inch triangle; three readings shall be made on the hinge, and three readings shall be made on the thrust washer. Any reading not within specified requirements shall constitute a defect.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.-

5.1.1.1 Type I intrenching tool (with carrier).- Each type I intrenching tool, with carrier, shall be completely folded and placed in the carrier. Forty intrenching tools shall then be packed flat in a snug fitting fiberboard box conforming to grade V3c or V3s, style RSC of PPP-B-636. Each box shall be closed, waterproofed by means of tape, and reinforced in accordance with the appendix of PPP-B-636.

5.1.1.2 Type II intrenching tool (without carrier).- Each type II intrenching tool, without carrier, shall be completely folded and completely wrapped in minimum 1/2 inch thick cushioning material conforming to type I or II, class B of PPP-C-843, and the wrap secured with tape, cord or twine. Alternatively, each tool shall be wrapped or sleeved in paperboard conforming to type III, style 1 or 2 of PPP-B-291. The wrap or sleeve shall be of sufficient bulk as to immobilize the tool within the box, or filler pads made from cushioning materials specified herein may be used to immobilize the wrapped tool. Forty wrapped intrenching tools shall then be packaged flat in a snug fitting fiberboard box conforming to grade V3c or V3s, style RSC of PPP-B-636. Each box shall be closed in accordance with method II of the appendix of PPP-B-636, except that top flaps shall be closed by the application of a 2 inch width tape conforming to PPP-T-76. The tape shall be centered over the seam formed by the closure of the outer flaps and shall extend down over the end panels not less than 2 inches.

5.1.2 Level C.- Intrenching tools shall be packaged to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing.- Packing shall be level A, B or C as specified (see 6.2).

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5.2.1 Level A.- One hundred twenty intrenching tools of one type only, packaged as specified in 5.1, shall be packed in a snug fitting cleated plywood shipping container conforming to overseas type of PPP-B-601. Each shipping container shall be provided with skids, and closed and strapped in accordance with PPP-B-601 and the appendix thereto.

5.2.2 Level B.- One hundred twenty intrenching tools of one type only, packaged as specified in 5.1, shall be packed in a snug fitting cleated plywood shipping container conforming to domestic type, style A or B of PPP-B-601. Each shipping container shall be provided with skids in accordance with PPP-B-601.

5.2.3 Level C.- Intrenching tools, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking.- In addition to any special marking required by the contract or order, interior boxes and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- The intrenching tool covered by this specification is intended to remove earth and shall also be capable of severing roots and clearing light brush.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type intrenching tool required (see 1.2).
(c) First article.- When preproduction sample is not required (see 3.1).
(d) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 This specification supersedes LP/P DES 21-69, dated 8 May 1969, and MIL-I-11519C; Intrenching , Tool, Hand; dated 17 September 1963.

Custodians:

Army - GL
Air Force - 84

Review activity:

Air Force - 84

User activities:

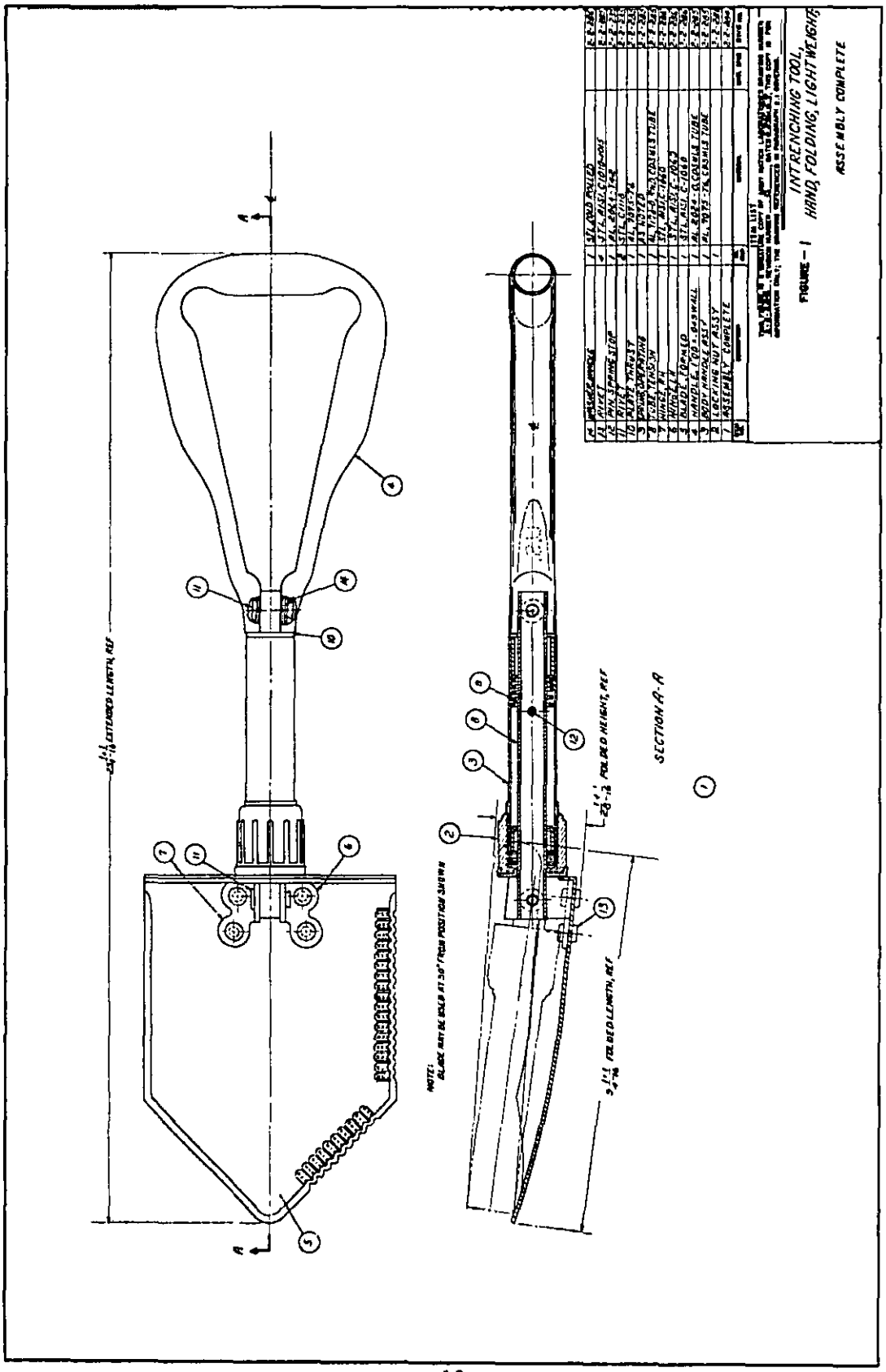
Navy - MC, YD

Preparing activity:

Army - GL

Project No. 5120-0620

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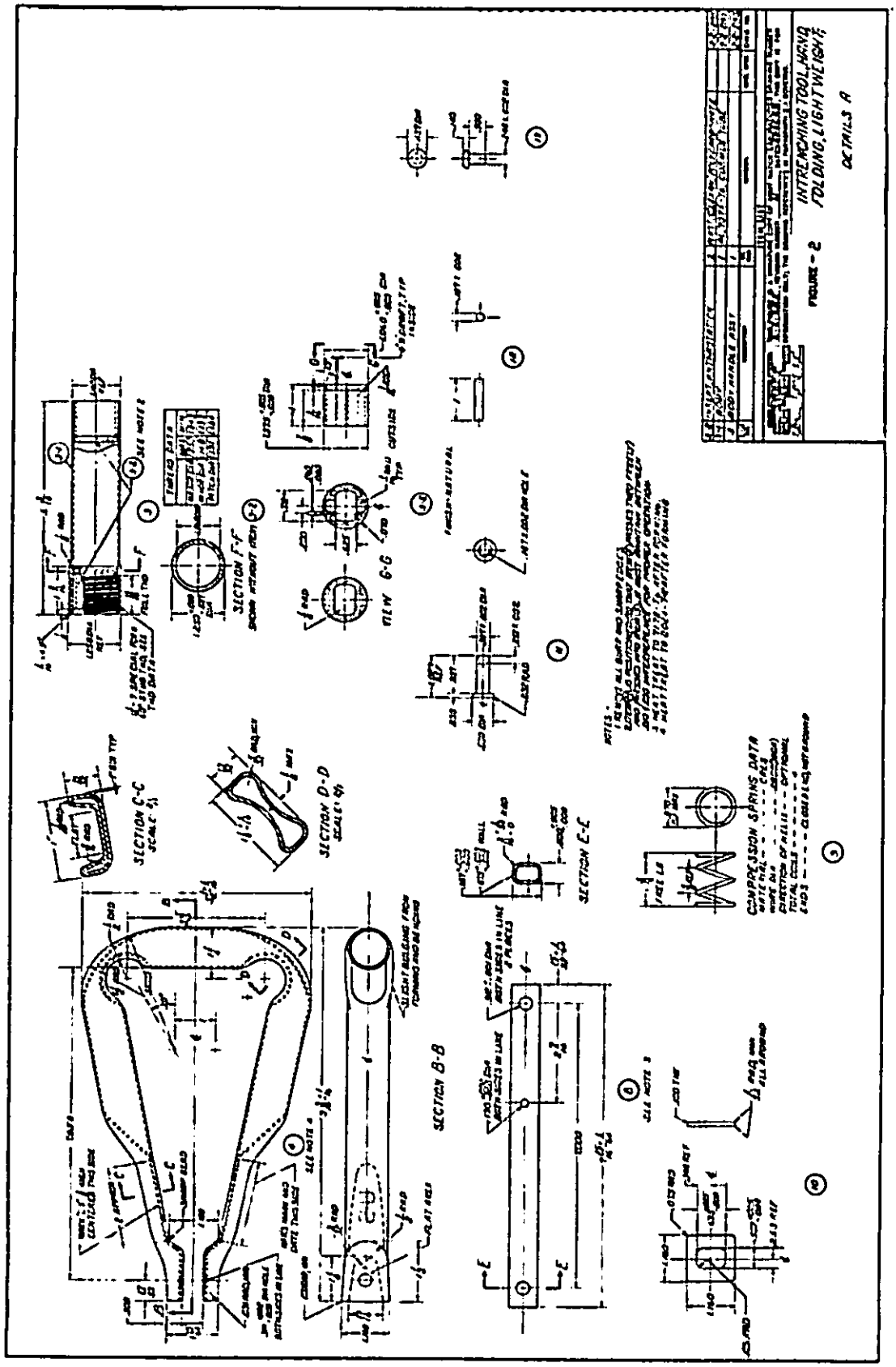
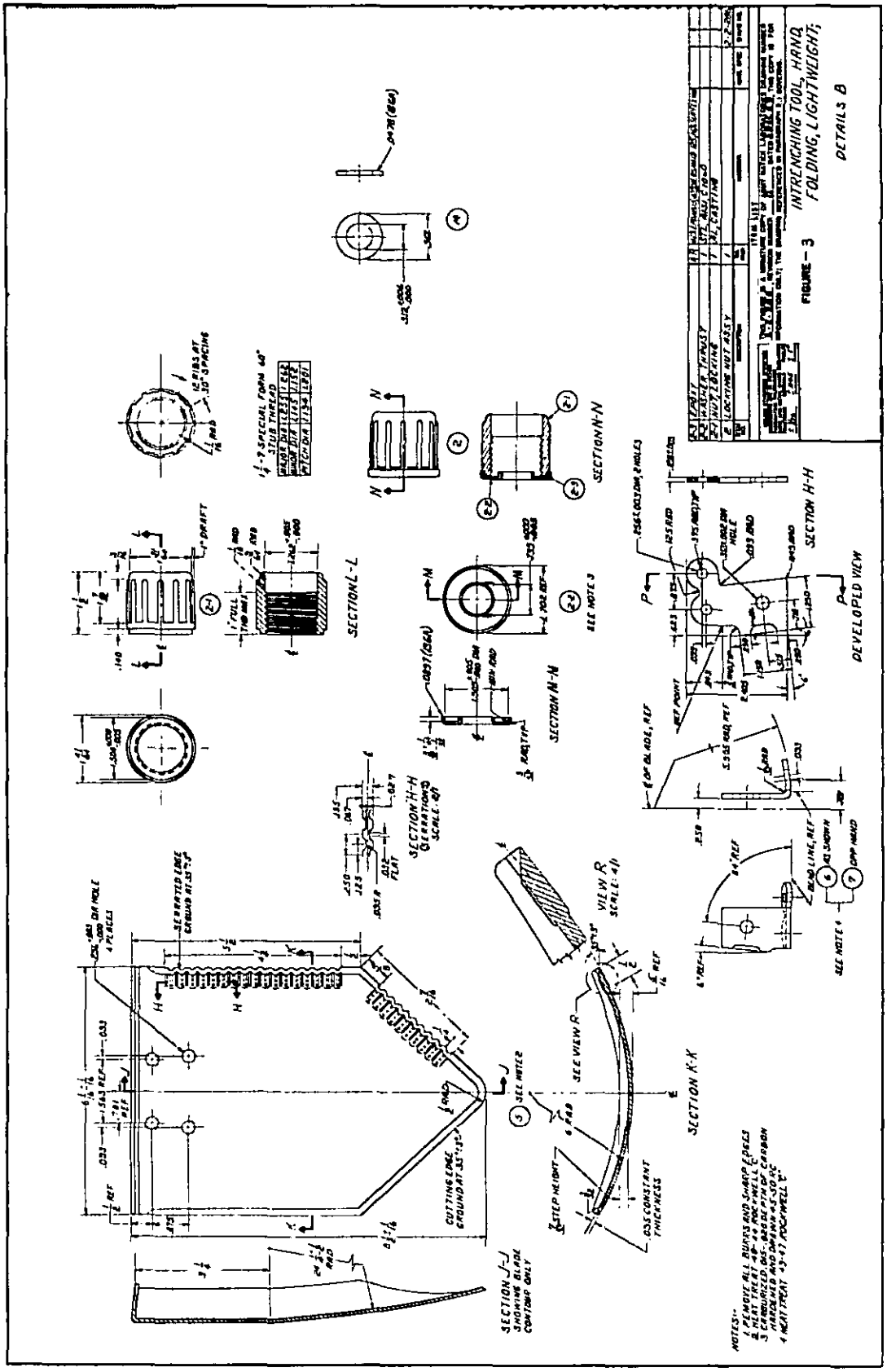


FIGURE 2 - INTERCHANGING TOOL ASSY, FOLDING, LIGHTWEIGHT

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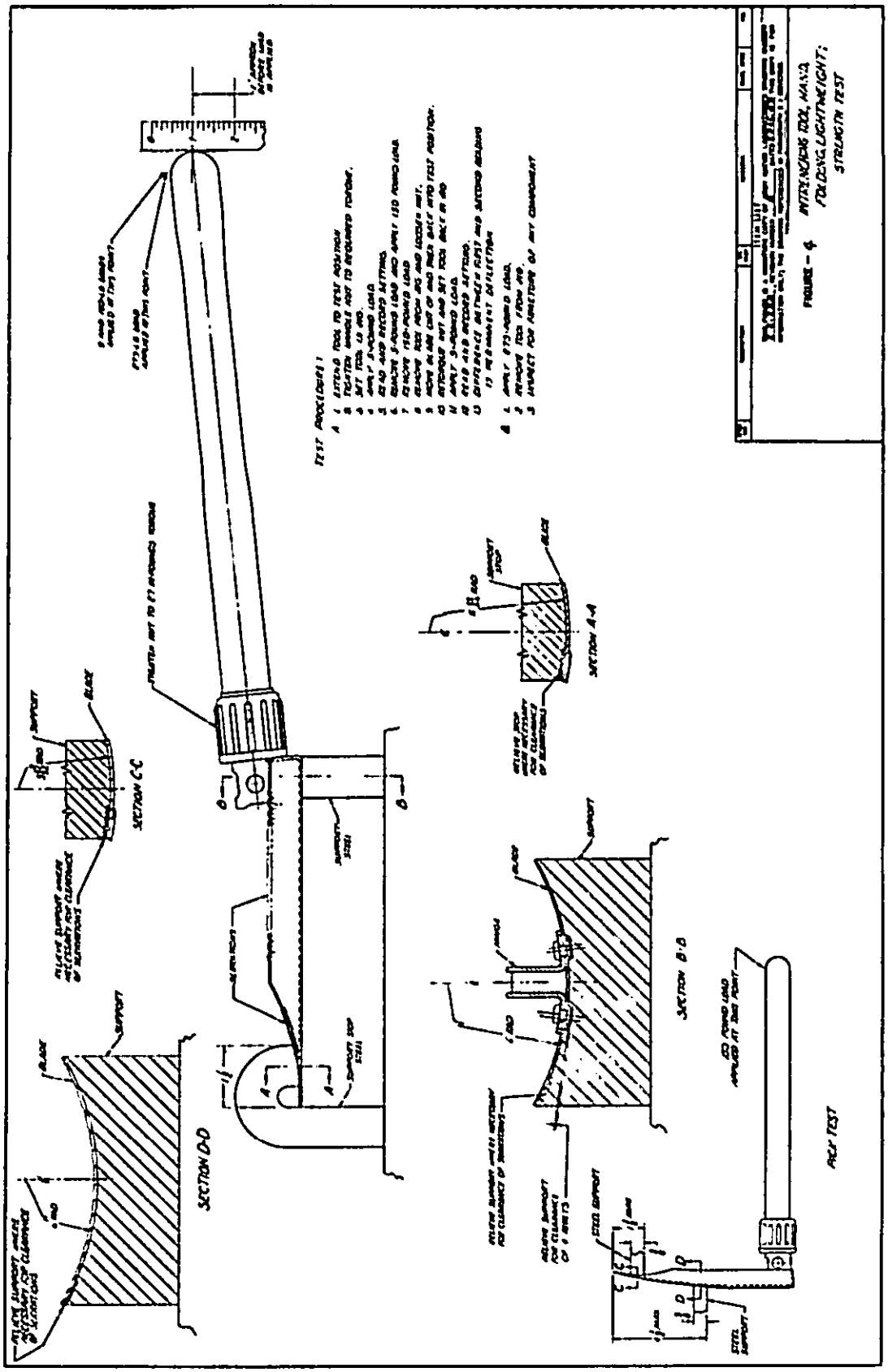


FIGURE - 4 INTENSIFIER TOOL, MAIN, FOLDING LIGHT WEIGHT; STRENGTH TEST

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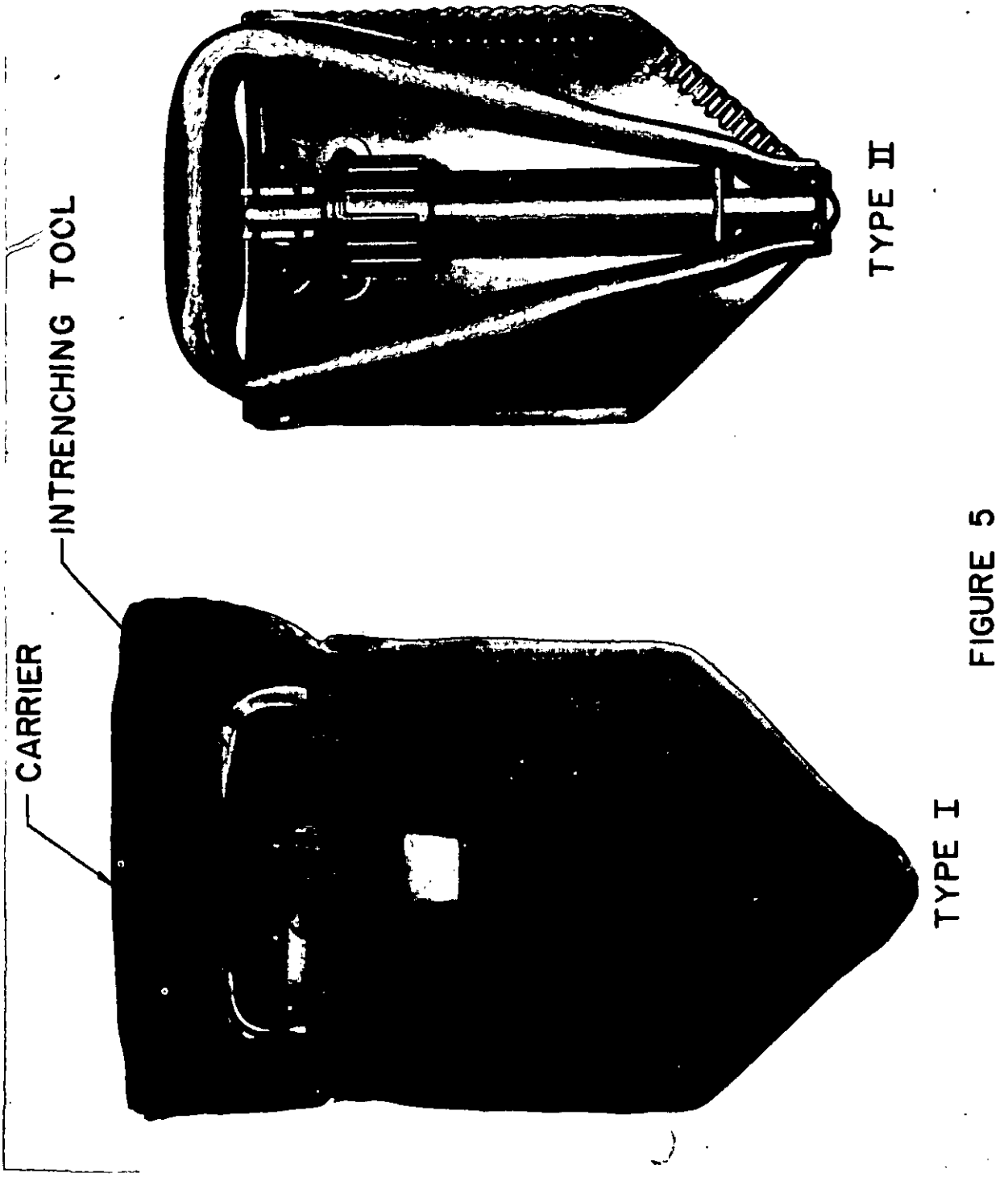


FIGURE 5

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION Intrenching Tool, Hand Folding, Lightweight, And Carrier MIL-I-43684

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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